Tema 1 ML

Tunsoiu Dan-Andrei

Setul de date “Bycicles”

1) Exploarea datelor:

Mai intai am vizualizat atributele setului de date si am afisat informatiile specifice pentru datele numerice.

This is for train data:

	data_ora	sezon	sarbatoare	zi_lucratoare	vreme	temperatura	temperatura_resimtita	umiditate	viteza_vant	ocazionali	inregistrati	total
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0000	3	13	16
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0000	8	32	40
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0000	5	27	32
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0000	3	10	13
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0000	0	1	1
...	...	...	...	...	...	...	...	...	...	...	...	...
6873	2012-12-12 19:00:00	4	0	1	2	12.30	14.395	56	15.0013	18	384	402
6874	2012-12-12 20:00:00	4	0	1	1	12.30	14.395	52	19.0012	25	243	268
6875	2012-12-12 21:00:00	4	0	1	2	12.30	14.395	52	16.9979	16	186	202
6876	2012-12-12 22:00:00	4	0	1	2	12.30	14.395	52	15.0013	4	118	122
6877	2012-12-12 23:00:00	4	0	1	2	11.48	13.635	56	15.0013	3	76	79

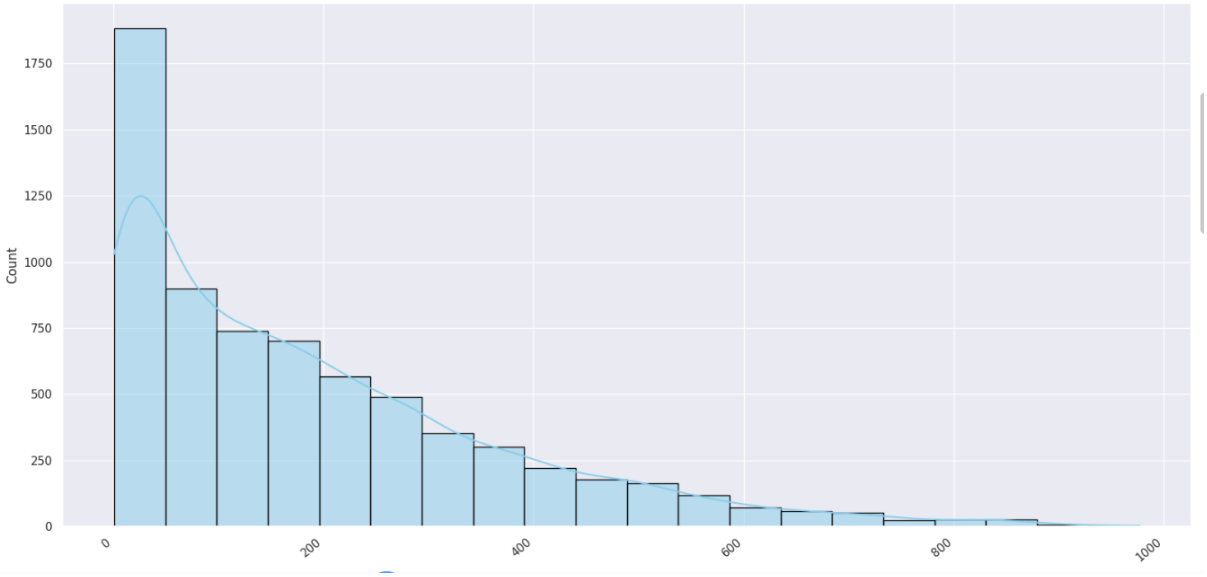
This is numerical data:

	sezon	sarbatoare	zi_lucratoare	vreme	temperatura	temperatura_resimtita	umiditate	viteza_vant	ocazionali	inregistrati	total
count	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000	6878.000000
mean	2.505670	0.031259	0.676069	1.418436	20.077839	23.505789	61.990259	12.576206	35.356208	153.511050	188.867258
std	1.116669	0.174030	0.468008	0.642373	8.168544	8.869723	19.827335	8.248823	49.064414	150.508752	179.669118
min	1.000000	0.000000	0.000000	1.000000	0.820000	0.760000	0.000000	0.000000	0.000000	0.000000	1.000000
25%	2.000000	0.000000	0.000000	1.000000	13.120000	15.910000	46.000000	7.001500	4.000000	35.000000	41.000000
50%	3.000000	0.000000	1.000000	1.000000	19.680000	23.485000	62.000000	11.001400	16.000000	115.000000	141.000000
75%	4.000000	0.000000	1.000000	2.000000	27.060000	31.060000	78.000000	16.997900	48.000000	220.000000	282.000000
max	4.000000	1.000000	1.000000	4.000000	41.000000	45.455000	100.000000	56.996900	362.000000	886.000000	977.000000

This is categorical data:

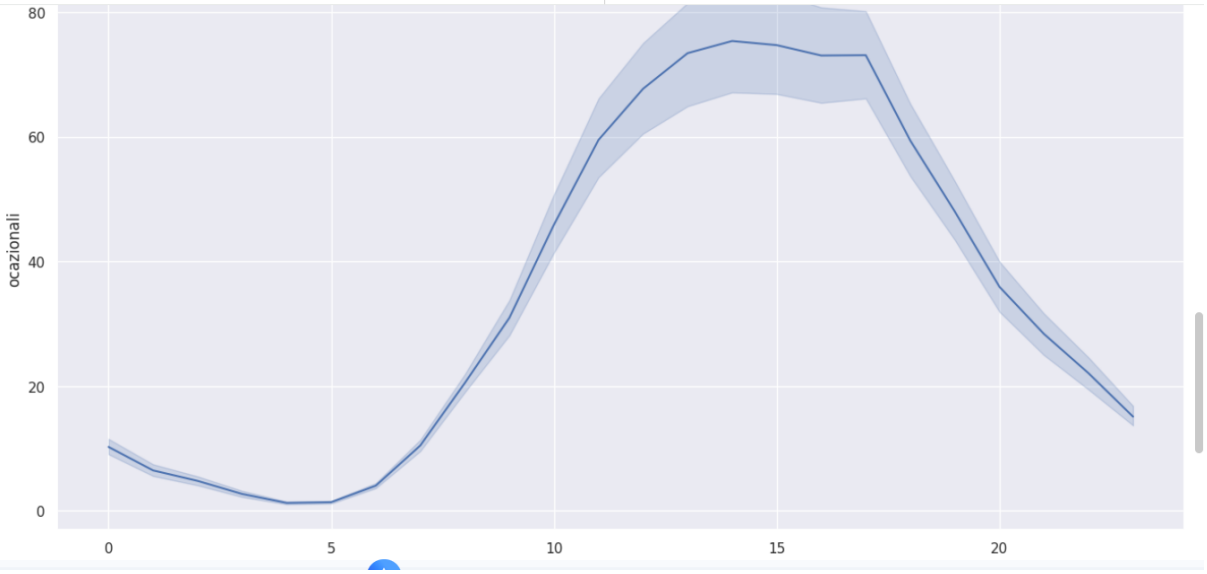
Dupa am afisat si o histograma cu distributia variabilei tinta (total):

```
Echilibru clase:
/tmp/ipython-input-2131598794.py:14: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLo
  graph.set_xticklabels(graph.get_xticklabels(), rotation=40, ha="right")
```

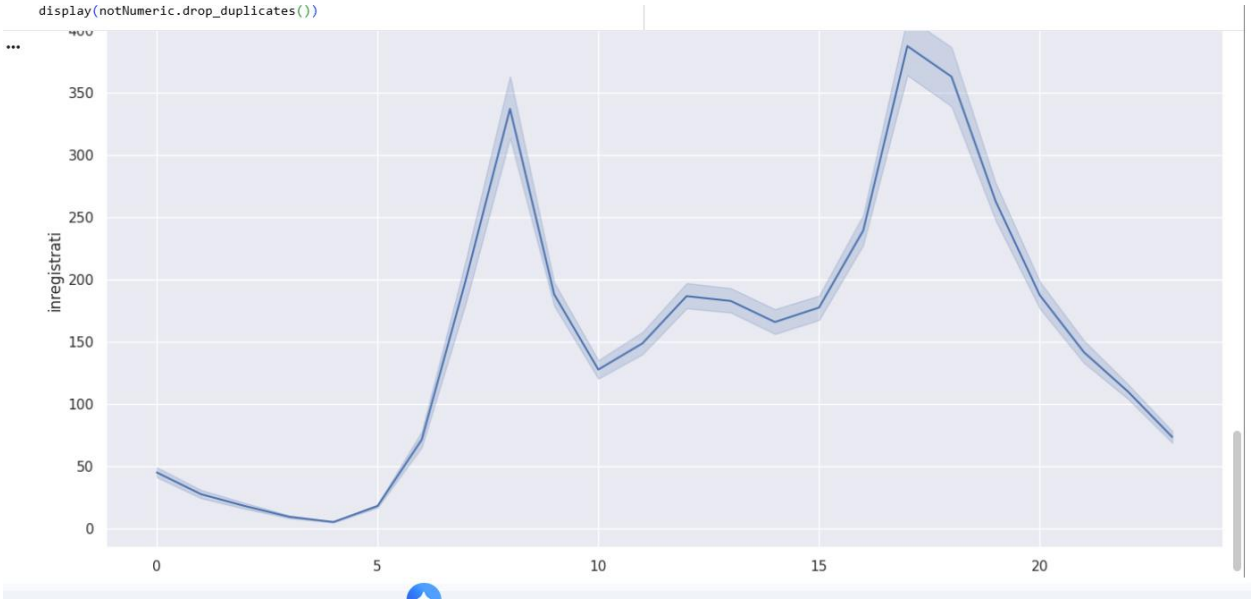


Dupa cum se poate observa aceasta are o distributie logaritmica, ceea ce era cat de cat de asteptat. A urmat sa transform sectiunea data_ora in date numerice pentru a putea utiliza modelele de antrenare si am facut o distributie a numarului de inregistrati si numarului de ocazionali in functie de ora.

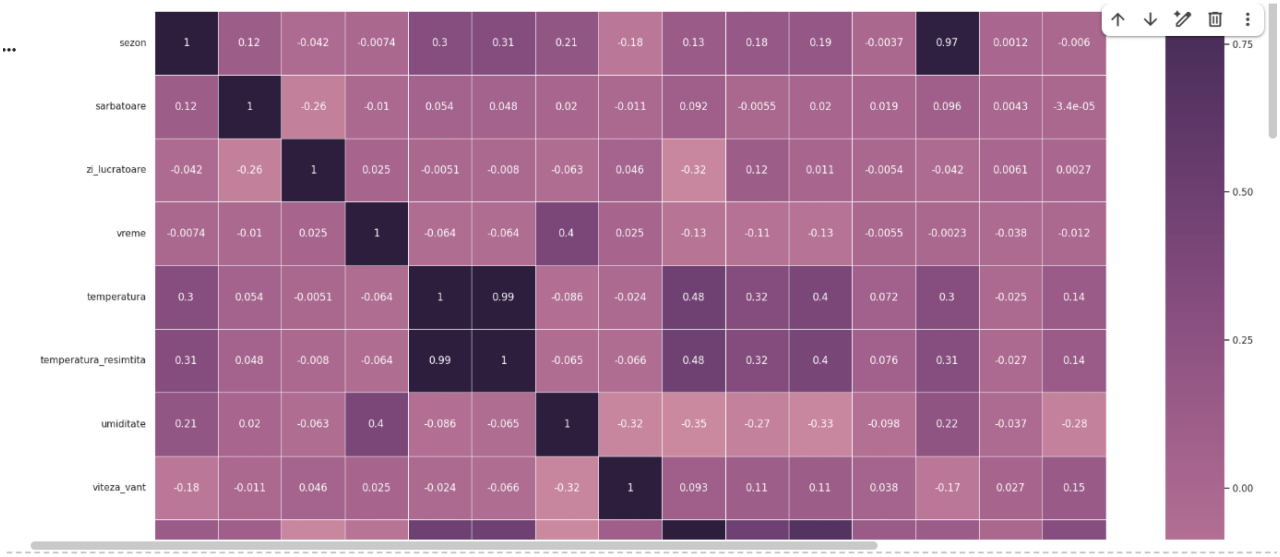
```
display(notNumeric.drop_duplicates())
```

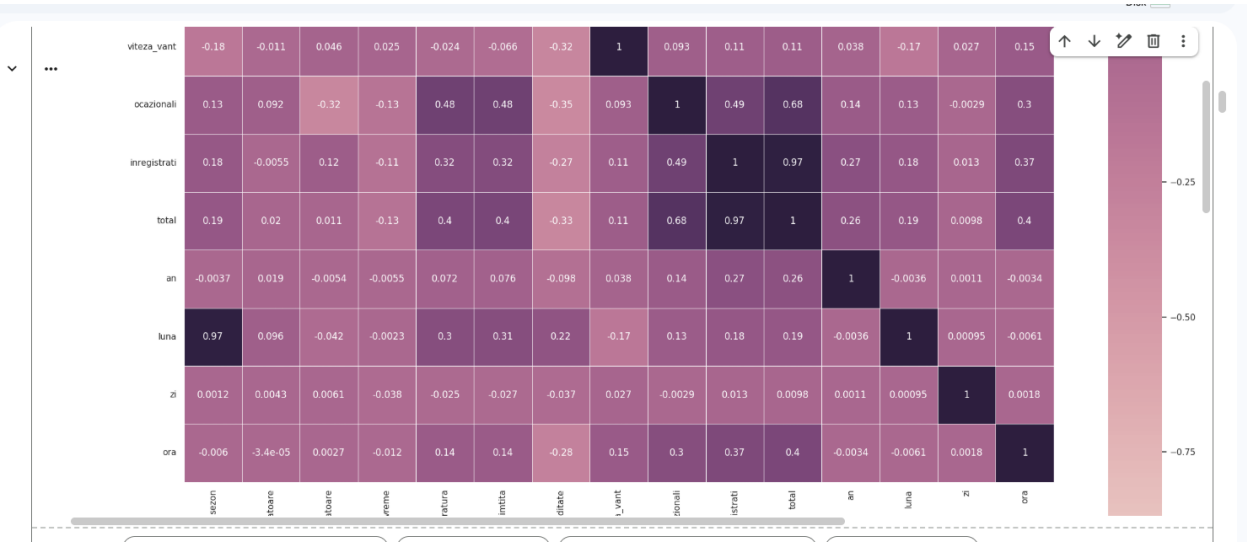


```
display(notNumeric.drop_duplicates())
```



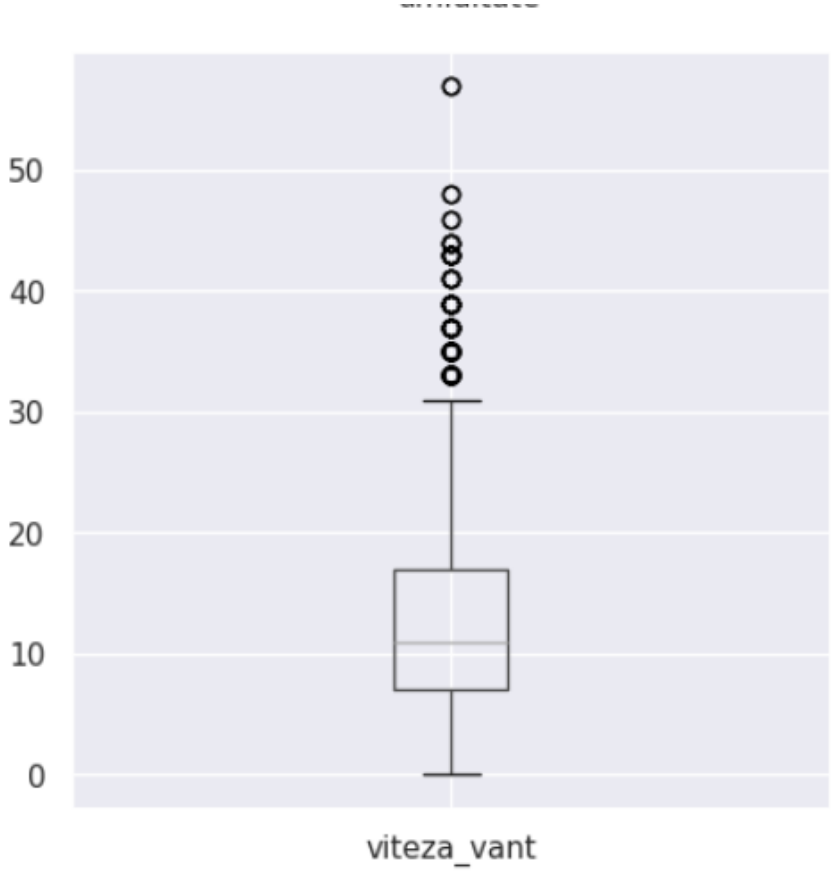
Se poate observa ca nu exista o relatie liniara intre ora si numarul de biciclisti ceea ce o sa duca la performante slabe pentru modelele liniare dar la performante bune pentru arbori de decizie in schimb. Am mai facut si matricia de corelatie intre atributele numerice in urma careia am scos coloanele pentru sezon si temperatura.





Am repetat procesul si pentru datele de training si dupa am analizat si fiecare coloane imparte individual pentru a observa daca exista sau nu outlieri si alte informatii importante legate de acestea.

Ex viteza vant:



2) Extragerea, standardizarea, selecția de atribute și suplimentarea valorilor lipsă:

Dupa care a urmat procesul de preprocesare al datelor, ceea ce la setul de date “bycicles” a fost destul de simplu datorita faptului ca toate datele erau numerice. Mai intai am scapat de outlieri, dupa unde aveam valori lipsa am folosit in imputer iterativ pentru a genera inlocuitori si dupa am standardizat datele ca sa aiba media 0 si deviatia standard corecta pentru modelele utilizate.

3) Utilizarea algoritmilor de invatare si cautarea parametrilor corecti:

Linear Regresionul a dat rezultate proaste, cum era de asteptat.

```
pd.DataFrame(results).set_index("model")
```

1 entry Filter

model	MAE	MSE	R2
Linear Regresion	106.00729548244895	20743.694640411515	0.3843235912659356

Show 25 per page

Mai interesant este SVR-ul care in functie de parametrii putea sa dea de la rezultate mult mai sable decat Linear Regresion pana la rezultate mai bune, chiar daca nu cu mult, cum se poate observa si in urmatorul tabel:

1 to 25 of 64 entries Filter

(kernel, degree, C, epsilon)	MAE	MSE	R2
rbf,3,1.8,0.3	94.2534576779773	21426.23749140856	0.36406560258101184
rbf,4,1.8,0.3	94.2534576779773	21426.23749140856	0.36406560258101184
rbf,3,1.8,0.1	94.26393546877996	21431.00952336549	0.36392396785544645
rbf,4,1.8,0.1	94.26393546877996	21431.00952336549	0.36392396785544645
rbf,3,1.4,0.1	96.01545701089096	22036.27504387525	0.3459595835710753
rbf,4,1.4,0.1	96.01545701089096	22036.27504387525	0.3459595835710753
rbf,3,1.4,0.3	96.01828436643653	22039.82497140149	0.3458542211142658
rbf,4,1.4,0.3	96.01828436643653	22039.82497140149	0.3458542211142658
linear,3,1.8,0.3	103.19531747733863	23087.20814212401	0.314767709270677
linear,4,1.8,0.3	103.19531747733863	23087.20814212401	0.314767709270677
linear,3,1.8,0.1	103.19751141514773	23092.11779754297	0.31462198985719425
linear,4,1.8,0.1	103.19751141514773	23092.11779754297	0.31462198985719425
linear,3,1.4,0.1	103.1991445301091	23101.778526642953	0.3143352577633636
linear,4,1.4,0.1	103.1991445301091	23101.778526642953	0.3143352577633636
linear,3,1.4,0.3	103.20274640258057	23107.6101645711	0.3141621737511431
linear,4,1.4,0.3	103.20274640258057	23107.6101645711	0.3141621737511431
rbf,3,1,0.1	98.65225688169117	23112.25520929739	0.31402430802832637
rbf,4,1,0.1	98.65225688169117	23112.25520929739	0.31402430802832637
rbf,3,1,0.3	98.65312836915996	23113.85302343302	0.31397688463119755
rbf,4,1,0.3	98.65312836915996	23113.85302343302	0.31397688463119755

1 to 25 of 64 entries Filter

(kernel, degree, C, epsilon)	MAE	MSE	R2
poly,4,0.4,0.1	133.28891098136282	34538.50201282918	-0.02510865354172398
poly,4,0.4,0.3	133.2856165516627	34536.229475014115	-0.025041204230261016
poly,4,1,0.3	128.0105107796757	32395.025692337615	0.03851009066384414
poly,4,1,0.1	127.99981383859699	32391.537703713973	0.03861361476341918
poly,4,1.4,0.3	125.33357869838997	31367.615150796253	0.06900381145355383
poly,4,1.4,0.1	125.31983269846408	31358.094101235787	0.06928639783473522
poly,4,1.8,0.3	123.16940961136882	30523.933287849704	0.0940444336039139
poly,4,1.8,0.1	123.1600353452637	30519.825880781744	0.09416634214897535
poly,3,0.4,0.1	119.39639428710821	29675.98082379199	0.11921180792648056
poly,3,0.4,0.3	119.40206079768043	29673.214454662426	0.11929391423591318
poly,3,1,0.1	114.3724188119416	27383.764858641855	0.18724517025322784
poly,3,1,0.3	114.37261928779677	27377.885740106296	0.1874196634980061
poly,3,1.4,0.3	113.3979232676642	26854.960651066725	0.20294016967773043
poly,3,1.4,0.1	113.39273001417969	26846.98473056636	0.20317689636384584
rbf,3,0.4,0.1	108.19661668392676	26480.124542549154	0.21406536955925803
rbf,4,0.4,0.1	108.19661668392676	26480.124542549154	0.21406536955925803
rbf,3,0.4,0.3	108.18883357063872	26469.305504364158	0.2143864804650617
rbf,4,0.4,0.3	108.18883357063872	26469.305504364158	0.2143864804650617
poly,3,1.8,0.3	112.54480473179323	26440.93404109657	0.21522855035270305
poly,3,1.8,0.1	112.54818102431001	26437.859282950478	0.21531980970094966
sigmoid,3,0.4,0.1	106.29062079631372	25018.294104831275	0.2574527472503536
sigmoid,4,0.4,0.1	106.29062079631372	25018.294104831275	0.2574527472503536
sigmoid,3,0.4,0.3	106.27801014928392	25018.073642112195	0.25745929062160433

De constatat ca SVR-ul a fost si algoritmul care a durat cel mai mult de rulat, ceea ce nu era o problema foarte mare la datasetul “Bycicles” dar la “Autovit” avea sa devina o adevarata provocare.

Dupa SVR a venit Random Forrest Regression care, dupa cum era de asteptat, a dat rezultate cu mult superioare fata de modelele liniare, dar din nou, in functie de parametrii acestea putea sa varieze destul de mult:

1 to 25 of 36 entries Filter

(tree, depth, percentage)	MAE	MSE	R2
100,6,0.2	60.30156352952849	8749.83761119917	0.7403033215223573
200,6,0.2	60.48333121543068	8862.688912933714	0.7369538755640885
150,6,0.2	60.628318175279176	8873.32888702392	0.7366380794098617
50,6,0.4	60.82287353532395	9042.397077591015	0.7316201064668105
50,6,0.2	61.239282894270026	9068.389575991478	0.7308486446637599
100,6,0.4	60.9870411157646	9089.697888699842	0.7302162102940932
200,6,0.4	60.90799134840898	9095.762717596253	0.7300362051340068
150,6,0.4	61.082762019293945	9126.02233670049	0.7291380944578372
50,6,0.6	61.33885441382187	9230.286051185745	0.7260435295595236
150,6,0.6	61.41177721575294	9271.122114687074	0.724831508202713
200,6,0.6	61.619491515886196	9317.89380547041	0.723443316411854
100,6,0.6	61.80242117589401	9361.607213423213	0.722145894979055
200,4,0.2	75.95171962820488	12828.576854794934	0.6192456423966913
150,4,0.2	75.6970238409665	12828.745105662814	0.6192406486821241
100,4,0.2	75.87926858857489	12848.692705244222	0.618648605110514
50,4,0.2	76.16116884338604	12918.282980691445	0.6165831492202918
150,4,0.4	76.6853070715024	13051.544020610849	0.6126279387380709
200,4,0.4	76.78202468535122	13083.803340040213	0.6116704766138559
150,4,0.6	77.11518871356036	13189.52073053255	0.6085327663627487
100,4,0.6	77.42993526757259	13215.742922310561	0.6077544872209235
100,4,0.4	77.4797815384352	13237.563171368625	0.6071068584927422
200,4,0.6	77.4396166631151	13248.208135502035	0.6067909141345953
50,4,0.4	77.63042631754571	13330.623073142538	0.6043448246740564

1 to 25 of 36 entries Filter

(tree, depth, percentage)	MAE	MSE	R2
50,2,0.4	98.53364470187589	20019.014664468723	0.40583221703411243
100,2,0.6	97.46071043293257	19565.656167106536	0.41928797486142133
200,2,0.6	97.38875427340878	19546.51329050201	0.41985613871675265
100,2,0.4	97.20022006560228	19537.69445734765	0.42011788319476506
150,2,0.4	97.23304074175203	19489.282906896853	0.4215547463013367
150,2,0.6	97.15532778392854	19440.35304106747	0.42300699309709255
200,2,0.2	96.95133869989618	19420.847244066867	0.42358592849192
50,2,0.6	96.89529827667015	19331.405068495602	0.42624058757747163
200,2,0.4	96.8278267232456	19267.399745523624	0.4281402765224912
100,2,0.2	96.49808889248555	19148.738310570192	0.4316621682295275
150,2,0.2	96.32055878262425	19128.713475288154	0.43225650876941946
50,2,0.2	95.69993930839966	18931.973022966224	0.43809579908091045
50,4,0.6	77.77485490808671	13404.356777507934	0.6021563957035533
50,4,0.4	77.63042631754571	13330.623073142538	0.6043448246740564
200,4,0.6	77.4396166631151	13248.208135502035	0.6067909141345953
100,4,0.4	77.4797815384352	13237.563171368625	0.6071068584927422
100,4,0.6	77.42993526757259	13215.742922310561	0.6077544872209235
150,4,0.6	77.11518871356036	13189.52073053255	0.6085327663627487
200,4,0.4	76.78202468535122	13083.803340040213	0.6116704766138559
150,4,0.4	76.6853070715024	13051.544020610849	0.6126279387380709
50,4,0.2	76.16116884338604	12918.282980691445	0.6165831492202918
100,4,0.2	75.87926858857489	12848.692705244222	0.6186486005110514
150,4,0.2	75.6970238409665	12828.745105662814	0.6192406486821241
200,4,0.2	75.95171962820488	12828.576854794934	0.619245642396691 3
100,6,0.6	61.80242117589401	9361.607213423213	0.722145894979055

Interesant este ca modelel care considerau un procent mai mic din numarul total de date per arbore s-au descurcat mai bine decat cele care considerau un procent mai mare ceea ce nu ma asteptam.

Dupa urmeaza Gradient Boosted Regression, atat in forma standard de prezicere a valorilor cat si in forma cu prezicerea de cuantile, si din nou avem rezultate foarte diferite in functie de ce parametrii folosim:

Gradiend Boosted Regressor (sklearn, squared_error):

1 to 25 of 48 entries Filter

(tree, depth, llr)	MAE	MSE	R2
150,6,0.05	34.334980937642044	2898.208541294032	0.9139806742531714
50,6,0.2	34.39185933231429	2899.889006250101	0.9139307977655403
100,6,0.05	35.134262053494034	2979.188490400032	0.9115771754980403
50,6,0.1	35.46788845387873	3006.131311905742	0.9107775079425083
200,4,0.1	36.72445170800007	3056.05687597621	0.9092957086524702
200,6,0.05	35.102216283437485	3060.579639913882	0.909161472244397
100,6,0.1	35.62439827705519	3113.379954889564	0.9075943498552674
100,6,0.2	36.35319558385019	3151.964310426375	0.906449159576389
150,4,0.1	37.44350105221321	3168.1736048549765	0.9059680649423371
50,4,0.3	37.56691807522325	3174.6380955861587	0.9057761977505633
200,4,0.05	39.01478290700833	3367.187777679395	0.900061290217018
200,6,0.2	37.80344455956725	3380.9210203405623	0.8996536852233816
150,6,0.2	37.799153219531085	3407.066522832303	0.898877682232704
100,4,0.1	39.341579071519554	3437.7612292895715	0.8979666580893992
50,4,0.2	38.869348307585284	3489.7033189224658	0.8964250079753919
150,4,0.3	39.40357800682652	3547.358763743269	0.8947137845011471
100,4,0.3	39.771328660634616	3623.55223412436	0.8924523492541286
50,6,0.3	38.50872421555524	3677.7584423417557	0.8908434997128631
150,4,0.05	40.93048455639946	3714.9465291798674	0.889739750928036
150,6,0.1	38.27765375252875	3788.8227403416436	0.8875470923314158
100,4,0.2	40.29175543424967	3810.3304745162104	0.8869087390984866

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Gradiend Boosted Regressor (sklearn, squared_error):

1 to 25 of 48 entries Filter

(tree, depth, llr)	MAE	MSE	R2
50,2,0.05	83.02861552244687	14350.111066594769	0.574086246468158
50,2,0.1	71.52752763067548	11034.664119943314	0.672489279526987 9
100,2,0.05	71.42707973748743	11019.628257684628	0.672935546493317 8
150,2,0.05	68.26776474233645	9872.860970188187	0.706971795939646 2
100,2,0.1	66.44905295795485	9154.144547052845	0.728303422439397 8
200,2,0.05	66.27863302034778	9133.732806186614	0.728909246403276
50,2,0.2	65.70396280690683	8927.035285038768	0.735044064222403 6
150,2,0.1	63.42038168203315	8185.877555784909	0.757041752530230 4
50,2,0.3	62.29773484155812	7761.440415240225	0.769639119535181 8
100,2,0.2	61.439003710410006	7569.989942743019	0.775321402339706 4
50,4,0.05	59.07049280350593	7566.620295883605	0.775421414035456 3
200,2,0.1	60.98960406525057	7544.853321804999	0.776067461553125 4
150,2,0.2	57.67519773355284	6690.251151661179	0.801432200290913 1
100,2,0.3	57.79585168454108	6688.808638994894	0.801475014313840 4
200,2,0.2	55.87862982489476	6210.942717768426	0.815658156677686 5
150,2,0.3	54.493859762698165	5870.932385592915	0.825749721554392 6
200,2,0.3	52.905907661180265	5587.989353977518	0.834147519179215 4
200,4,0.2	44.02839741832087	4743.468400764852	0.859213045672345 9
100,4,0.05	45.54191523665503	4664.038409389415	0.861570541416644 8
200,6,0.1	40.11926766632902	4436.141561362946	0.868334558887351 6
50,4,0.1	44.56931720721253	4391.220909430225	0.869667811528183 1
150,4,0.2	41.70602083965002	4221.38628079808	0.874708532385653 6
50,6,0.05	43.53543501451886	4217.880963734634	0.87481257079628 9
200,6,0.3	41.91234370376421	4194.701619981461	0.875500537687730 2
150,6,0.3	41.50703706456814	4121.462936397238	0.877674274356688 8

Nimic surprinzator la forma normala, ne asteptam ca modelele cu un numar mai mare de arbori si adancime mai mare sa se descurce mai bine decat cele mai putin complexe.

1 to 25 of 144 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.50,0.1,6,200	12.323856553493417	16.240131614404472	20.15640667531553
q 0.50,0.2,6,150	12.945802082275675	16.51383204053441	20.081861998793144
q 0.50,0.2,6,200	13.046703011721208	16.74495682939896	20.44321064707671
q 0.50,0.1,6,150	12.827477612875171	16.791559128697564	20.755640644519957
q 0.50,0.2,6,100	12.964084619022369	16.948692872712815	20.933301126403258
q 0.50,0.05,6,200	12.681286348550012	16.961490916882937	21.24169548521586
q 0.50,0.3,6,150	13.535475297173951	17.046075871293386	20.556676445412823
q 0.50,0.3,6,100	12.972214462300359	17.067585324641257	21.16295618698216
q 0.50,0.2,6,50	13.134446336582604	17.11058852166114	21.086730706739676
q 0.50,0.3,6,200	13.73538895674832	17.160877096192728	20.58636523563714
q 0.50,0.1,6,100	12.764171392471361	17.178238977020698	21.592306561570037
q 0.50,0.3,6,50	13.48171104144355	17.34535654101121	21.209002040578866
q 0.50,0.05,6,150	13.039044974262286	17.86781714177428	22.696589309286278
q 0.50,0.1,4,200	13.584852514098607	18.00778767349434	22.43072283289008
q 0.50,0.3,4,150	14.242072768023274	18.093942723160616	21.945812678297955
q 0.50,0.2,4,200	14.195140868329542	18.445348024352356	22.69555518037517
q 0.50,0.3,4,200	14.265799298415049	18.468301024703432	22.670802750991818
q 0.50,0.3,4,100	14.41052903404996	18.790929155798985	23.171329277548015
q 0.50,0.2,4,150	14.356239202214823	18.90502585077591	23.45381249933699
q 0.50,0.2,4,100	14.065677991686385	19.10647164773868	24.147265303790977
q 0.50,0.1,6,50	13.311778379664185	19.135121626945992	24.958464874227797
q 0.50,0.05,6,100	13.466976009174846	19.200248555820727	24.923521102487642

76 to 100 of 144 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.95,0.05,6,150	69.1579538057213	37.49234763412003	5.826741462518765
q 0.50,0.05,2,100	22.46918062879796	37.1577708347971	51.84636104079624 6
q 0.05,0.2,6,150	4.72113619899952	37.15030292521808	69.57946965143663
q 0.05,0.2,6,200	4.792655422528905	37.02903369657293	69.26541197061695
q 0.50,0.1,2,50	22.32387599205906	36.6966689093561	51.06946182665314
q 0.95,0.1,6,100	66.55020826045138	36.141266137176125	5.732324013900883
q 0.95,0.3,4,150	66.01431166559884	36.08458914594545	6.154866626292071
q 0.05,0.3,6,150	4.774636005686997	35.90650263195568	67.03836925822436
q 0.95,0.3,4,100	65.33229570945242	35.58105144587645	5.829807182300482
q 0.95,0.2,4,150	64.99269788240522	35.42700322294732	5.861308563489418
q 0.95,0.2,4,200	65.05425165536664	35.40422015122177	5.754188647076882
q 0.95,0.3,4,200	64.29247324427895	35.286735180331405	6.280997116383864
q 0.05,0.3,6,200	4.977303628520927	34.98906302600319 4	65.00082242348546
q 0.95,0.05,6,200	64.15056560447637	34.98369911110194	5.816832617727527
q 0.50,0.05,2,150	21.44260423146807	34.77437224016907	48.10614024887007
q 0.95,0.2,6,50	62.78080090721018	34.39186816106323 6	6.002935414916277
q 0.50,0.2,2,50	21.43824584302829	33.53604140245433	45.63383696188038
q 0.50,0.05,4,50	20.543551207129653	33.50265728505252	46.46176336297538
q 0.50,0.05,2,200	21.470292564827652	33.49733257670645	45.52437258858524
q 0.50,0.1,2,100	21.437865556828328	33.47714017348724	45.516414790146165
q 0.95,0.1,6,150	60.074509710594725	32.91913505566309	5.763760400731454
q 0.95,0.3,6,50	57.781019892607816	32.06566054418608	6.350301195764335
q 0.95,0.3,6,100	56.885909495510084	31.816914287459443	6.747919079408804
q 0.50,0.3,2,50	21.467018196281597	31.754953978447084	42.04288976061257 4

Acelasi fenomen se poate observa si la cuantile.

Ultima este Quantile Regressorul care are urmatorile valori in functie de alpha:

```
We are at: 0.5, 2
We are at: 0.95, 2
```

1 to 25 of 27 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.50,0.4	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,0.6	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,0.8	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,1	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,1.2	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,1.4	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,1.6	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,1.8	44.947841816367266	69.79653193612775	94.64522205588823
q 0.50,2	44.947841816367266	69.79653193612775	94.64522205588823
q 0.05,0.4	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,0.6	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,0.8	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,1	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,1.2	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,1.4	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,1.6	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,1.8	9.667602295409182	96.16629241516966	182.66498253493015
q 0.05,2	9.667602295409182	96.16629241516966	182.66498253493015
q 0.95,0.4	346.2562749500998	185.7049650698603	25.153655189620775

Cu cat alphaul creste cu atat modelul pare ca se descurca mai prost si nu ajunge la performantele la Gradient Boosted Regressor.

Setul de date “Autovit”:

1) Exploarea datelor:

Pentru vizualizarea datelor am facut ca la setul de date “Bycicles”, mai intai am afisat un tabel pentru a imi face o idee despre cu ce o sa lucrez, dupa am afisat o descriere a datelor unde am observat mai multe lucruri interesante:

1) Multe coloane aveau date lipsa
2) Cateva coloane aveau foarte multe clase distincte sau aveau atat de putine exemple incat nu mai erau semnificative si am scapat de acestea. (Versiune, nume, Garantie de la producator, etc.)
3) Aveam foarte multe date categorice care trebuiau preprocesate

This is numerical data:

	pret	Anul fabricației	Km	Putere
count	18988.000000	18988.000000	1.894500e+04	18983.000000
mean	25424.654675	2016.609490	1.381373e+05	179.185587
std	29824.752654	4.780368	9.048320e+04	93.524671
min	600.000000	1969.000000	1.000000e+00	3.000000
25%	9490.000000	2014.000000	6.543000e+04	116.000000
50%	16500.000000	2018.000000	1.400000e+05	150.000000
75%	29999.000000	2020.000000	2.030000e+05	200.000000
max	666666.000000	2024.000000	2.080288e+06	1020.000000

	Capacitate cilindrica	Consum Extraurban	Consum Urban	Emisii CO2
count	18447.000000	8352.000000	13064.000000	9635.000000
mean	1977.705210	4.879095	6.752013	155.590939
std	659.091494	1.173531	2.260320	1753.493092
min	479.000000	2.900000	2.200000	2.000000
25%	1560.000000	4.100000	5.300000	113.000000
50%	1968.000000	4.700000	6.300000	132.000000
75%	1998.000000	5.400000	7.700000	156.000000
max	6750.000000	12.900000	75.000000	172139.000000

	Numar de portiere	Numar locuri
count	18759.000000	14757.000000
mean	4.707021	5.000745
std	0.625786	0.676494
min	2.000000	1.000000
25%	5.000000	5.000000
50%	5.000000	5.000000
75%	5.000000	5.000000
max	6.000000	9.000000

```
max            6.000000      9.000000

This is categorical data:
                    nume Oferit de Are VIN (Serie sasiu)          Marca  \
count              18988     18988                10877          18988
unique              8188         2                    1             66
top     Volkswagen Passat     Firma                   Da  Mercedes-Benz
freq                 119     13058                10877           2431

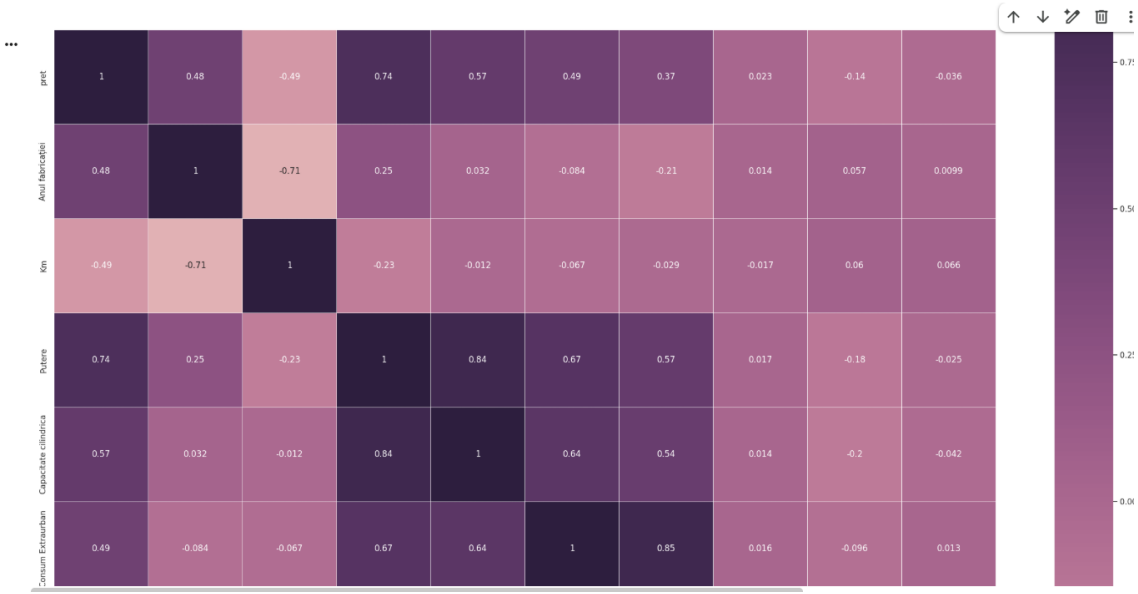
          Model   Versiune Combustibil Transmisie Cutie de viteze  \
count     18988      14946       18988      16872           18987
unique      622       6837           8          4               2
top     Octavia  xDrive30d      Diesel       Fata        Automata
freq        515        125       11759       8513           12286

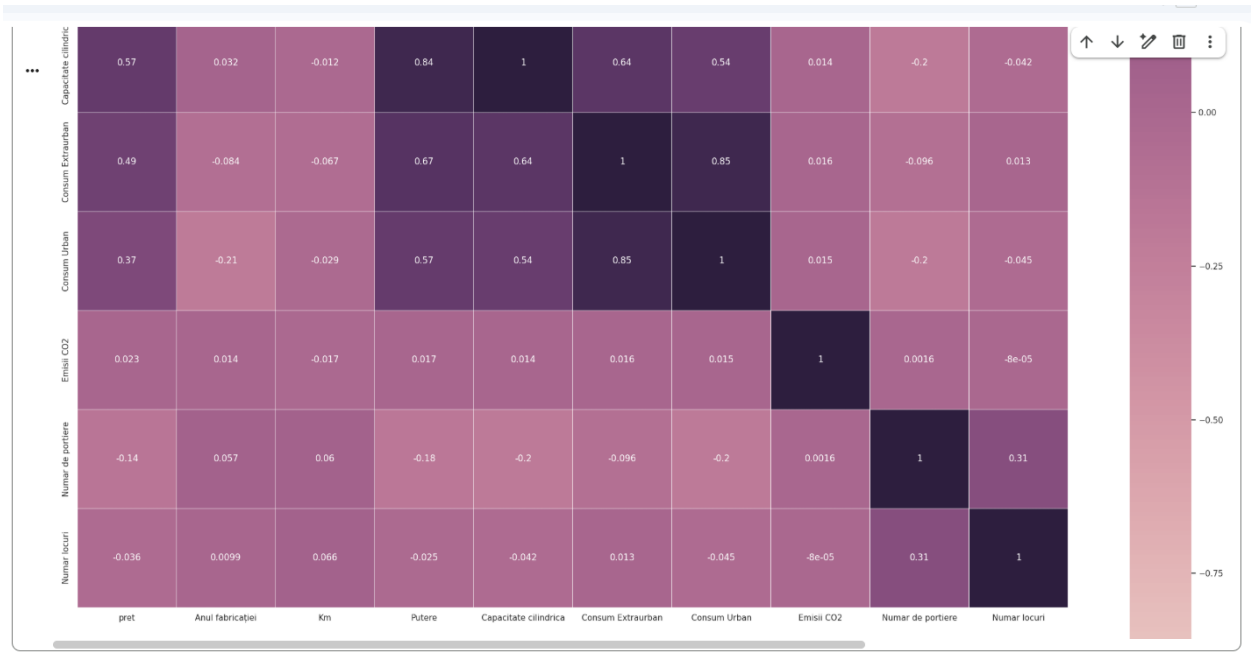
       Tip Caroserie  ... Electronice si sisteme de asistenta  \
count          18988  ...                              18678
unique             9  ...                              13745
top              SUV  ...                  ['Servodirectie']
freq            7652  ...                                123

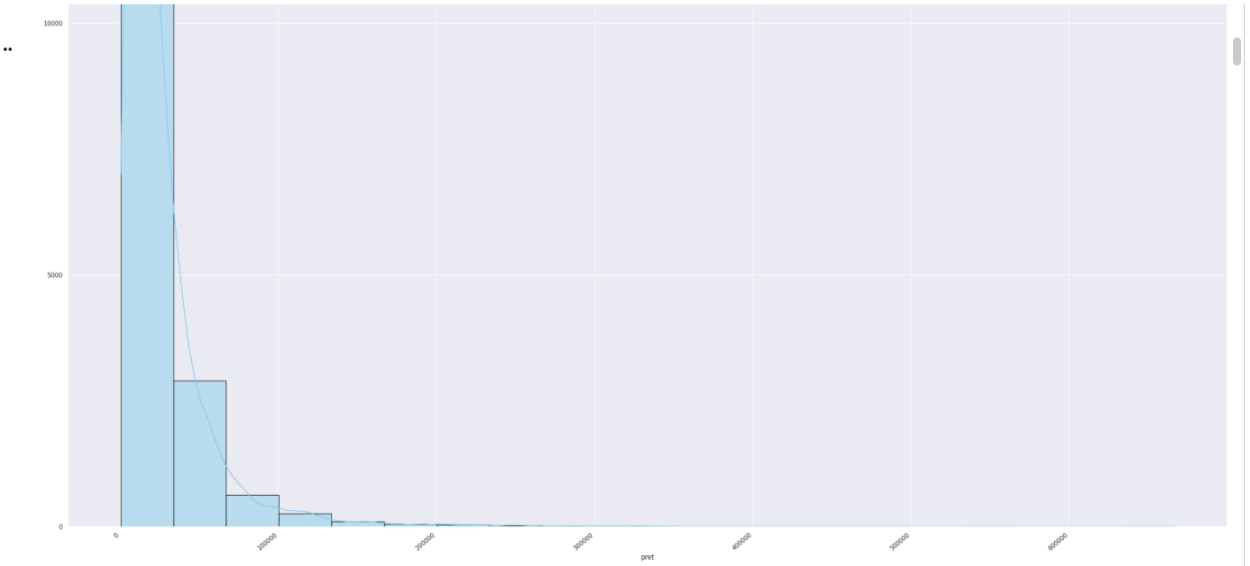
                                          Siguranta Norma de poluare  \
count                                        18474            12914
unique                                        7470               13
top     ['ABS', 'ESP', 'EBD', 'Franare asistata', 'Air...     Euro 6
freq                                           449             8042

       Tara de origine Data primei inmatriculari  \
count            11275                      7390
unique              29                      3183
top           Germania                2019-08-06
freq              4594                        27
```

Dupa aceea am facut o histograma cu distributia pretului si apoi matricea de corelatie:







Dupa cum se poate observa, la autovit sunt si mai logaritmic distribuite preturile si exista mai multi outlieri puternici. Am mai facut si testul Chi pentru a mai scapa din coloanele de care nu aveam nevoie.

2) Extragerea, standardizarea, selecția de atribute și suplimentarea valorilor lipsă:

Preprocesarea datelor la autovit a fost mult mai dificila decat la “Bycicles” din mai multe motive, primul dintre acestea este faptul ca avem coloane care contin liste de atribute care a trebuit sa le sparg in mai multe coloane care sa contina 1 daca acea masina are acel atribut si 0 daca nu. O alta problema a fost numarul mare de date categorice si

numarul de categorii diferite in acestea ceea ce a dus la a avea 324 de coloane in setul nostru de date ceea ce in final o sa duca la un timp crescut de antrenare a modelelor noastre.

	Anul fabricației	Km	Putere	Capacitate cilindrica	Consum Extraurban	Consum Urban	Emisii CO2	Numar de portiere	Numar locuri	Oferit de_Firma	Oferit de_Privat	Marca_Abarth	Marca_Aixam	Marca_Alfa Romeo	Marca_Altul	Marca_Aston Martin	Marca_Audi	Ma
0	0.498833	-0.505364	1.388140	1.611043	0.756008	0.405390	0.708739	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	3.122354	-(
1	-0.145723	0.211517	-1.065197	-0.813265	-1.590429	-1.392369	-1.595246	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
2	0.069129	-0.442728	-0.952544	-0.653898	-1.101617	-1.173840	-1.085149	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
3	-0.790280	0.763763	-0.877442	-0.592727	-1.285313	-0.791414	-0.976985	-4.025541	0.0	-1.480318	1.480318	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
4	0.283981	-1.367328	0.086369	-0.752094	1.374409	1.503145	1.058587	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4741	0.498833	-0.869510	-0.739755	-0.813265	-0.380271	-0.391913	-0.493459	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
4742	0.069129	0.932897	0.311676	-0.560532	0.981894	1.175351	0.847248	0.509473	0.0	-1.480318	1.480318	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
4743	0.498833	0.294122	-0.689687	-0.753704	-0.903688	-0.354355	-0.978076	-1.758034	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
4744	1.358241	-1.492488	-0.289142	0.839965	-3.364337	-3.751592	-3.415900	0.509473	0.0	0.675531	-0.675531	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(
4745	-1.434836	1.426273	-0.189005	-0.927558	0.482597	1.011454	0.379699	0.509473	0.0	-1.480318	1.480318	-0.015902	-0.024294	-0.047754	-0.012983	-0.011244	-0.320271	-(

4746 rows × 324 columns

3) Utilizarea algoritmilor de invatare si cautarea parametrilor corecti:

Regresia liniara a dat rezultate mult mai bune la autovit decat la bycicles, ceea ce era de asteptat deoarece intuitiv vorbind ar trebui sa avem o relatie liniara intre atributele unei masini si pretul acesteia:

```
results.append(metrics)
pd.DataFrame(results).set_index("model")
```

1 entry Filter

model	MAE	MSE	R2
Linear Regresion	7469.63998083786	179815893.01452905	0.7905953938663873

Show 25 per page

SVR:

SVR-ul a fost algoritmul care a durat cel mai mult de rulat, ajungand la trei, trei ore si jumatate, pentru o singura rulare si nici nu a adus rezultate superbe:

1 to 25 of 64 entries Filter

(kernel, degree, C, epsilon)	MAE	MSE	R2
linear,3,1.8,0.1	6890.426548345706	343856956.2333862	0.5995613664664383
linear,4,1.8,0.1	6890.426548345706	343856956.2333862	0.5995613664664383
linear,3,1.8,0.3	6890.422585255874	343857977.92960304	0.599560176650064
linear,4,1.8,0.3	6890.422585255874	343857977.92960304	0.599560176650064
linear,3,1.4,0.1	6997.204359401035	359194925.1862058	0.5816995340465685
linear,4,1.4,0.1	6997.204359401035	359194925.1862058	0.5816995340465685
linear,3,1.4,0.3	6997.206296084561	359195101.92478275	0.5816993282256508
linear,4,1.4,0.3	6997.206296084561	359195101.92478275	0.5816993282256508
linear,3,1,0.3	7164.184547156299	379974420.34386325	0.5575007720450751
linear,4,1,0.3	7164.184547156299	379974420.34386325	0.5575007720450751
linear,3,1,0.1	7164.184381278843	379975425.72996765	0.55749901222643
linear,4,1,0.1	7164.184381278843	379975425.72996765	0.55749901222643
linear,3,0.4,0.3	7830.614945237845	444721320.63766074	0.4820997664548707
linear,4,0.4,0.3	7830.614945237845	444721320.63766074	0.4820997664548707
linear,3,0.4,0.1	7830.6243301409295	444722701.45197475	0.4820981584275069
linear,4,0.4,0.1	7830.6243301409295	444722701.45197475	0.4820981584275069
sigmoid,3,1.8,0.1	14859.283048150866	884470862.7938571	-0.0300105822405762 9
sigmoid,4,1.8,0.1	14859.283048150866	884470862.7938571	-0.0300105822405762 9
sigmoid,3,1.8,0.3	14859.283048150866	884470862.7938572	-0.0300105822405765 1
sigmoid,4,1.8,0.3	14859.283048150866	884470862.7938572	-0.0300105822405765 1
sigmoid,3,1.4,0.3	15084.262882486508	895849512.5958651	-0.0432615893689021 2

Interesant de observat este faptul ca aici kernelul cel mai bun este cel liniar fate de la “Bycicles” unde cel mai bun era “rbf”.

1 to 25 of 64 entries Filter

(kernel, degree, C, epsilon)	MAE	MSE	R2
poly,3,0.4,0.1	15948.569198639212	938069337.8105574	-0.09242868868307808
poly,3,0.4,0.3	15948.565346327374	938065687.7968923	-0.09242443805950362
rbf,3,0.4,0.3	15866.63684606738	935426807.6073672	-0.08935133001851048
rbf,4,0.4,0.3	15866.63684606738	935426807.6073672	-0.08935133001851048
rbf,3,0.4,0.1	15866.63684606738	935426807.6073669	-0.08935133001851026
rbf,4,0.4,0.1	15866.63684606738	935426807.6073669	-0.08935133001851026
poly,3,1,0.1	15891.5260269945	930870756.92621	-0.08404558099700776
poly,3,1,0.3	15891.522654076523	930867160.9185667	-0.08404139326614835
poly,4,0.4,0.3	15920.42745060133	927902119.9785638	-0.08058845470881404
poly,4,0.4,0.1	15920.425817899968	927900465.8145208	-0.08058652835200486
rbf,3,1,0.1	15689.600049929639	927487761.2604676	-0.08010591324539673
rbf,3,1,0.3	15689.600049929639	927487761.2604676	-0.08010591324539673
rbf,4,1,0.1	15689.600049929639	927487761.2604676	-0.08010591324539673
rbf,4,1,0.3	15689.600049929639	927487761.2604676	-0.08010591324539673
sigmoid,3,0.4,0.3	15711.607677597145	927118863.9725908	-0.07967631389254426
sigmoid,4,0.4,0.3	15711.607677597145	927118863.9725908	-0.07967631389254426
sigmoid,3,0.4,0.1	15711.60525015028	927118818.9989135	-0.07967626151844653
sigmoid,4,0.4,0.1	15711.60525015028	927118818.9989135	-0.07967626151844653
poly,3,1.4,0.3	15854.130650937943	926498983.2797027	-0.07895443180432649
poly,3,1.4,0.1	15854.126916643516	926495356.9938763	-0.07895020981300763
rbf,3,1.4,0.1	15575.0416648967	922459771.2638433	-0.0742505672732705
rbf,3,1.4,0.3	15575.0416648967	922459771.2638433	-0.0742505672732705
rbf,4,1.4,0.1	15575.0416648967	922459771.2638433	-0.0742505672732705
rbf,4,1.4,0.3	15575.0416648967	922459771.2638433	-0.0742505672732705

Urmeaza Random Forrest care a dat rezultate mai bune decat SVR-ul si a durat si mult mai putin de antrenat (in jur de 40 de min):

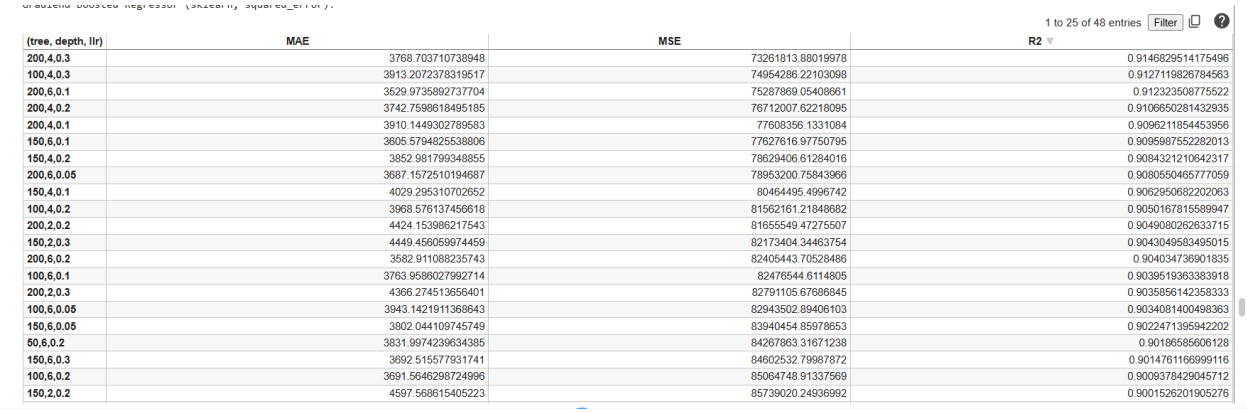
1 to 25 of 36 entries Filter

(tree, depth, percentage)	MAE	MSE	R2
200,6,0.6	5844.796293475658	149436950.05749163	0.8259731932256459
150,6,0.6	5876.727577972697	150820050.59642863	0.8243625034320807
50,6,0.6	5848.842662511653	152593335.07207343	0.822297424917829
100,6,0.4	5812.549020803124	154850365.4361658	0.819669000107 87
100,6,0.6	5922.68791649027	156320606.53357288	0.8179568307730988
200,6,0.4	5848.519344444769	156483079.09670824	0.8177676233425402
150,6,0.4	5814.130439086486	158090690.68463394	0.8158954791969824
50,6,0.4	5826.371994578172	159795212.50524613	0.8139104782356791
200,6,0.2	5695.0344367509215	161505405.53357974	0.811918872869174
150,6,0.2	5728.233408941966	163469834.71573716	0.8096311967785305
100,6,0.2	5817.1321585038795	165075873.75868776	0.8077608839402926
50,6,0.2	5759.624212857055	165823155.56165838	0.8068906368837143
100,4,0.4	7270.225142967287	200673209.10022858	0.7663060054997135
150,4,0.4	7270.424519959974	201518638.1061211	0.7653214611136476
50,4,0.4	7397.172078552832	202289694.0881248	0.764423527835828 9
200,4,0.4	7287.881282548681	202303524.26936346	0.76440742189765 8
200,4,0.6	7319.735202704531	202879436.40189633	0.7637367433983394
100,4,0.6	7366.547732674435	203661943.9300192	0.7628254742219829
150,4,0.6	7404.83468173848	204857534.47815126	0.7614331491964723
50,4,0.6	7451.477097248071	206548317.34277895	0.7594641479369663
200,4,0.2	7096.221377322125	211514710.25669387	0.7536805348502238
100,4,0.2	7219.737430157195	211638361.96001506	0.7535365362536723
150,4,0.2	7188.756070242767	212024214.88656488	0.7530871912110562
50,4,0.2	7305.504065064751	232515867.89214084	0.7292236358004531

1 to 25 of 36 entries Filter

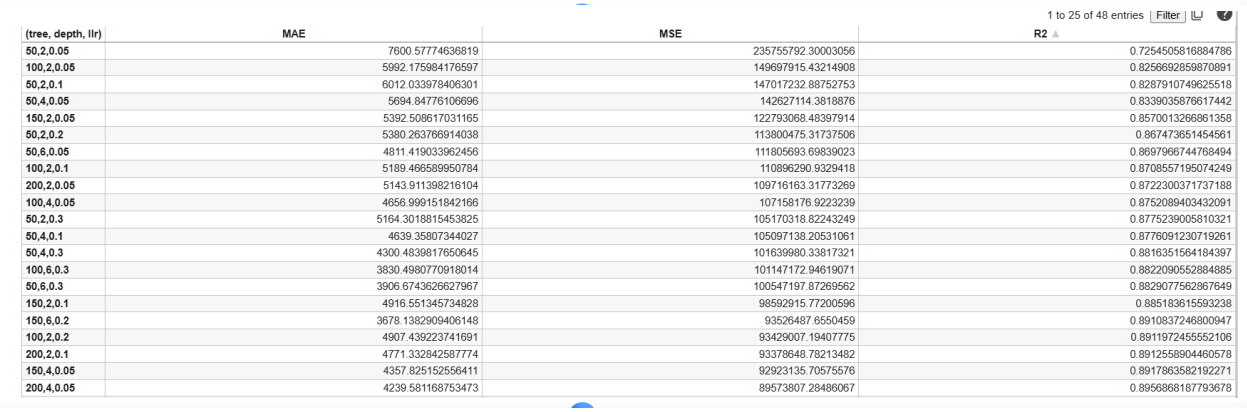
(tree, depth, percentage)	MAE	MSE	R2
50,2,0.6	10235.379216051062	383541514.36271006	0.553346712548244
150,2,0.6	10215.64675108409	383235540.86802083	0.5537030339949669
100,2,0.4	10478.430271020314	380547372.55472785	0.5568335405226656
200,2,0.6	10130.041001244328	380195359.3430832	0.5572434775237979
50,2,0.4	10156.536356523731	379248172.3799987	0.5583465241433956
150,2,0.4	10340.366536604073	378623666.8491027	0.5590737920869191
200,2,0.4	10172.893755309053	373840120.3577482	0.5646444713642118
100,2,0.6	10129.230369918945	373665633.34721905	0.5648476702200123
200,2,0.2	10169.327817364108	367793564.4468484	0.5716859883166903
150,2,0.2	10039.739382532387	367514145.5949846	0.5720113855529407
50,2,0.2	10107.661298343328	367459327.2462704	0.5720752242633513
100,2,0.2	9946.155049554505	367101107.1975938	0.5724923894368226
50,4,0.2	7305.504065064751	232515867.89214084	0.7292236358004531
150,4,0.2	7188.756070242767	212024214.88656488	0.7530871912110562
100,4,0.2	7219.737430157195	211638361.96001506	0.7535365362536723
200,4,0.2	7096.221377322125	211514710.25669387	0.7536805348502238
50,4,0.6	7451.477097248071	206548317.34277895	0.7594641479369663
150,4,0.6	7404.83468173848	204857534.47815126	0.7614331491964723
100,4,0.6	7366.547732674435	203661943.9300192	0.7628254742219829
200,4,0.6	7319.735202704531	202879436.40189633	0.7637367433983394
200,4,0.4	7287.881282548681	202303524.26936346	0.76440742189765 8
50,4,0.4	7397.172078552832	202289694.0881248	0.764423527835828 9
150,4,0.4	7270.424519959974	201518638.1061211	0.7653214611136476
100,4,0.4	7270.225142967287	200673209.10022858	0.7663060054997135

Gradient Boosted Regression insa este algoritmul care a adus cele mai bune rezultate, atat pentru cuantile cat si pentru cazul de prezicere al pretului dupa cum se poate observa din tabelul de mai jos:



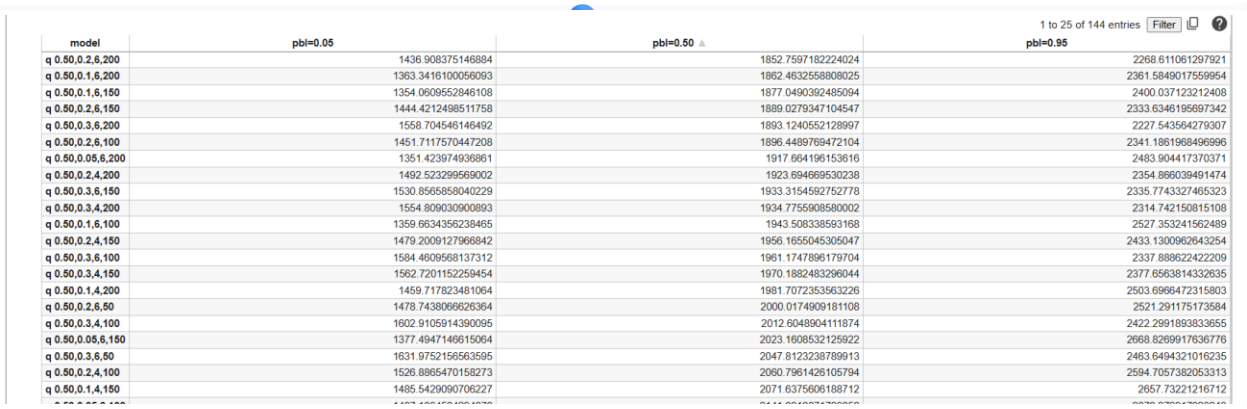
1 to 25 of 48 entries Filter

(tree, depth, llr)	MAE	MSE	R2
200,4,0.3	3768.703710738948	73261813.88019978	0.9146829514175496
100,4,0.3	3913.2072378319517	74954286.22103098	0.9127119826784563
200,6,0.1	3529.9735892737704	75287869.05408661	0.912323508775522
200,4,0.2	3742.7598618495185	76712007.62218095	0.910665028143293 5
200,4,0.1	3910.1449302789583	77608356.1331084	0.9096211854453956
150,6,0.1	3605.5794825538806	77627616.97750795	0.9095987552282013
150,4,0.2	3852.981799348855	78629406.61284016	0.9084321210642317
200,6,0.05	3687.1572510194687	78953200.75843966	0.9080550465777059
150,4,0.1	4029.295310702652	80464495.4996742	0.9062950682202063
100,4,0.2	3968.576137456618	81562161.21848682	0.9050167815589947
200,2,0.2	4424.153986217543	81655549.47275507	0.9049080262633715
150,2,0.3	4449.4560599744459	82173404.34463754	0.9043049583495015
200,6,0.2	3582.911088235743	82405443.70528486	0.904034736901835
100,6,0.1	3763.9586027992714	82476544.6114805	0.9039519363383918
200,2,0.3	4366.274513656401	82791105.67686845	0.9035856142358333
100,6,0.05	3943.1421911368643	82943502.89406103	0.9034081400498363
150,6,0.05	3802.044109745749	83940454.85978653	0.9022471395942202
50,6,0.2	3831.9974239634385	84267863.31671238	0.90186585606128
150,6,0.3	3692.515577931741	84602532.79987872	0.901476116699911 6
100,6,0.2	3691.5646298724996	85064748.91337569	0.900937842904571 2
150,2,0.2	4597.568615405223	85739020.24936992	0.900152620190527 6



1 to 25 of 48 entries Filter

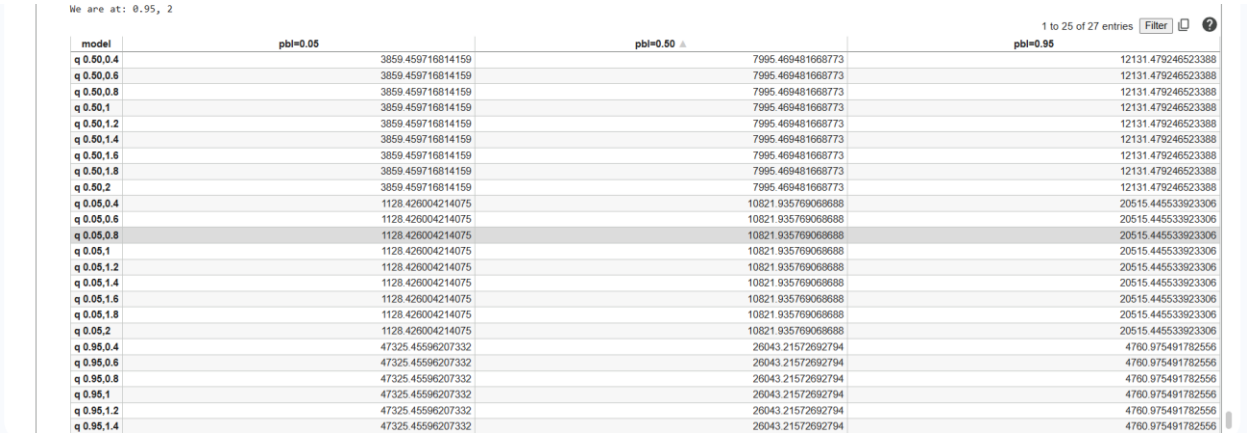
(tree, depth, llr)	MAE	MSE	R2
50,2,0.05	7600.57774636819	235755792.30003056	0.7254505816884786
100,2,0.05	5992.175984176597	149697915.43214908	0.8256692859870891
50,2,0.1	6012.033978406301	147017232.88752753	0.8287910749625518
50,4,0.05	5694.84776106696	142627114.3818876	0.8339035876617442
150,2,0.05	5392.508617031165	122793068.48397914	0.8570013266861358
50,2,0.2	5380.263766914038	113800475.31737506	0.867473651454561
50,6,0.05	4811.413033962456	111805693.69839023	0.8697966744768494
100,2,0.1	5189.466589950784	110896290.9329418	0.8708557195074249
200,2,0.05	5143.911398216104	109716163.31773269	0.8722300371737188
100,4,0.05	4656.999151842166	107158176.9223239	0.8752089403432091
50,2,0.3	5164.3018815453825	105170318.82243249	0.8775239005810321
50,4,0.1	4639.35807344027	105097138.20531061	0.8776091230719261
50,4,0.3	4300.4839817650645	101639980.33817321	0.8816351564184397
100,6,0.3	3830.4980770918014	101147172.94619071	0.8822090552884885
50,6,0.3	3906.6743626627967	100547197.87269562	0.8829077562867649
150,2,0.1	4916.55134573482 8	98592915.77200596	0.885183615593238
150,6,0.2	3678.1382909406148	93526487.6550459	0.8910837246800947
100,2,0.2	4907.439223741691	93429007.19407775	0.8911972455552106
200,2,0.1	4771.332842587774	93378648.78213482	0.8912558904460578
150,4,0.05	4357.825152556411	92923135.70575576	0.8917863582192271
200,4,0.05	4239.581168753473	89573807.28486067	0.8956868187793678



1 to 25 of 144 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.50,0.2,6,200	1436.908375146884	1852.7597182224024	2268.611061297921
q 0.50,0.1,6,200	1363.3416100056093	1862.4632558808025	2361.5849017559954
q 0.50,0.1,6,150	1354.0609552846108	1877.0490392485094	2400.037123212408
q 0.50,0.2,6,150	1444.4212498511758	1889.027934710454 7	2333.6346195697342
q 0.50,0.3,6,200	1558.704546146492	1893.1240552128997	2227.543564279307
q 0.50,0.2,6,100	1451.7117570447208	1896.4489769472104	2341.1861968496996
q 0.50,0.05,6,200	1351.423974936861	1917.664196153616	2483.904417370371
q 0.50,0.2,4,200	1492.523299569002	1923.694669530238	2354.866039491474
q 0.50,0.3,6,150	1530.8565858040229	1933.3154592752778	2335.7743327465323
q 0.50,0.3,4,200	1554.809030900893	1934.7755908580002	2314.742150815108
q 0.50,0.1,6,100	1359.6634356238465	1943.508338593168	2527.353241562489
q 0.50,0.2,4,150	1479.2009127966842	1956.1655045305047	2433.1300962643254
q 0.50,0.3,6,100	1584.4609568137312	1961.1747896179704	2337.888622422209
q 0.50,0.3,4,150	1562.7201152259454	1970.1882483296044	2377.6563814332635
q 0.50,0.1,4,200	1459.717823481064	1981.7072353563226	2503.6966472315803
q 0.50,0.2,6,50	1478.7438066626364	2000.0174909181108	2521.291175173584
q 0.50,0.3,4,100	1602.9105914390095	2012.6048904111874	2422.2991893833655
q 0.50,0.05,6,150	1377.4947146615064	2023.1608532125922	2668.8269917636776
q 0.50,0.3,6,50	1631.9752156563595	2047.8123238789913	2463.6494321016235
q 0.50,0.2,4,100	1526.8865470158273	2060.7961426105794	2594.7057382053313
q 0.50,0.1,4,150	1485.5429090706227	2071.6375606188712	2657.73221216712

Ultimul dintre modele este cel de Quantile Regressor care are urmatoarele rezultate:



```
We are at: 0.95, 2
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1 to 25 of 27 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.50,0.4	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,0.6	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,0.8	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.2	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.4	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.6	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.8	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,2	3859.459716814159	7995.469481668773	12131.479246523388
q 0.05,0.4	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,0.6	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,0.8	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1.2	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1.4	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1.6	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1.8	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,2	1128.426004214075	10821.935769068688	20515.445533923306
q 0.95,0.4	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,0.6	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,0.8	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.2	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.4	47325.45596207332	26043.21572692794	4760.975491782556

We are at: 0.95, 2

1 to 25 of 27 entries Filter

model	pbl=0.05	pbl=0.50	pbl=0.95
q 0.95,0.4	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,0.6	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,0.8	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.2	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.4	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.6	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,1.8	47325.45596207332	26043.21572692794	4760.975491782556
q 0.95,2	47325.45596207332	26043.21572692794	4760.975491782556
q 0.50,0.4	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,0.6	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,0.8	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.2	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.4	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.6	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,1.8	3859.459716814159	7995.469481668773	12131.479246523388
q 0.50,2	3859.459716814159	7995.469481668773	12131.479246523388
q 0.05,0.4	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,0.6	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,0.8	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1	1128.426004214075	10821.935769068688	20515.445533923306
q 0.05,1.2	1128.426004214075	10821.935769068688	20515.445533923306

We are at: 0.95, 2

model	pbl=0.05	pbl=0.50
q 0.05,0.4	1128.426004214075	10821.935769068688
q 0.05,0.6	1128.426004214075	10821.935769068688
q 0.05,0.8	1128.426004214075	10821.935769068688
q 0.05,1	1128.426004214075	10821.935769068688
q 0.05,1.2	1128.426004214075	10821.935769068688
q 0.05,1.4	1128.426004214075	10821.935769068688
q 0.05,1.6	1128.426004214075	10821.935769068688
q 0.05,1.8	1128.426004214075	10821.935769068688
q 0.05,2	1128.426004214075	10821.935769068688
q 0.50,0.4	3859.459716814159	7995.469481668773
q 0.50,0.6	3859.459716814159	7995.469481668773
q 0.50,0.8	3859.459716814159	7995.469481668773
q 0.50,1	3859.459716814159	7995.469481668773
q 0.50,1.2	3859.459716814159	7995.469481668773
q 0.50,1.4	3859.459716814159	7995.469481668773
q 0.50,1.6	3859.459716814159	7995.469481668773
q 0.50,1.8	3859.459716814159	7995.469481668773
q 0.50,2	3859.459716814159	7995.469481668773
q 0.95,0.4	47325.45596207332	26043.21572692794
q 0.95,0.6	47325.45596207332	26043.21572692794
q 0.95,0.8	47325.45596207332	26043.21572692794
q 0.95,1	47325.45596207332	26043.21572692794
q 0.95,1.2	47325.45596207332	26043.21572692794